

JEYARAAMAN K

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PROFESSIONAL SUMMARY

Python developer with hands-on experience in Django (MVT), REST APIs, and SQL, currently building full-stack web applications as part of a professional internship. Comfortable across the stack — from responsive frontends with HTML5/CSS3/JavaScript to backend logic and database design. Brings additional strength in applying computer vision and machine learning (OpenCV, YOLOv8-OB, TensorFlow) to real-world engineering problems.

EDUCATION

B. S. Abdur Rahman Crescent Institute Of Science And Technology, Chennai
B.Tech in Electronics and Communication Engineering

Sep 2022 – Apr 2026
CGPA: 7.4/10

WORK EXPERIENCE

Python Full-Stack + AI Intern
Careerschool HR Solutions

Feb 2026 – Present

- Build and maintain full-stack web applications using Python and Django following the MVT architecture.
- Design responsive, cross-browser user interfaces with HTML5, CSS3, JavaScript (ES6+), and CSS Grid/Flexbox, backed by SQL-driven data models.
- Integrate computer vision workflows into web applications by connecting OpenCV pipelines to Django backends through REST APIs.

Computer Vision Intern
Arkin Labs Pvt. Limited

Oct 2025 – Nov 2025

- Developed a drone-based depth estimation system for automated collision avoidance during flight.
- Applied YOLOv8-OB for real-time overhead wire detection and used stereo image disparity to calculate precise distance to obstacles.
- Trained and deployed real-time vision-based detection models, handling the full workflow from model training through live deployment.

Internet of Things (IoT) Intern
Denvik Technology Pvt. Limited

Jun 2024 – Jul 2024

- Completed structured training on embedded systems and microcontrollers, with focus on ATMEGA 4808/4809 architecture.
- Gained practical experience in PCB design and hardware-software integration for embedded applications.
- Worked with data communication protocols and storage technologies, including JSON, MQTT, and MariaDB.

PROJECTS

Fall Detection Using Machine Learning
Technologies: Python, TensorFlow, Keras, CNN-GRU

- Designed a wearable AI-based fall detection system using a hybrid CNN-GRU model to classify fall events from accelerometer and gyroscope data.
- Built a smartwatch prototype with an ESP32-C3 Mini, MPU6050 motion sensor, MAX30102 heart rate sensor, and a 0.96-inch OLED display, housed in a custom 3D-printed enclosure.
- Enabled Bluetooth connectivity with an Android application to automatically send emergency SMS alerts, share live GPS location, and trigger emergency calls upon detecting a fall.

Depth Estimation for Autonomous Drones
Technologies: Python, PyTorch, NumPy, Ultralytics YOLOv8-OB, OpenCV, Stereo Vision

- Implemented a real-time overhead wire detection model using Ultralytics YOLOv8-OB, trained on annotated aerial imagery.
- Applied stereo vision and image disparity calculations to estimate real-world distance between the drone and detected obstacles.
- Delivered a collision avoidance module to support safer autonomous navigation for drone flight systems.

Text Recognition Using Optical Character Recognition (OCR)

Technologies: Python, OpenCV, Tesseract OCR

- Built a real-time text recognition tool that extracts printed text from live camera and video feeds using Tesseract OCR.
- Applied image preprocessing techniques, including noise reduction and thresholding, to improve OCR accuracy under varying lighting conditions.
- Delivered a lightweight, camera-driven pipeline suitable for real-time document and signage text extraction.

TECHNICAL SKILLS

Programming Languages:	Python, JavaScript (ES6+)
Web Technologies:	Django (MVT), HTML5, CSS3, REST APIs
Databases & Data Formats:	SQL, JSON
Libraries & Frameworks:	Django, OpenCV, TensorFlow, Keras, PyTorch, NumPy, Tkinter
AI & Computer Vision:	Machine Learning, Computer Vision, Image Processing, YOLOv8-OBB, Tesseract OCR
Tools & IDEs:	Google Colab, PyCharm, Visual Studio Code
Version Control & OS:	Git, GitHub, Windows
Concepts:	Object-Oriented Programming (OOP), Networking Fundamentals
Professional Skills:	Problem Solving, Team Collaboration, Communication, Time Management

CERTIFICATIONS

- NPTEL – Data Analytics with Python
- Spoken Tutorial, IIT Bombay – Python Programming
- TechIgnite: Redefining Energy with Innovation – Siemens
- NPTEL – Introduction to Industry 4.0 and Industrial Internet of Things (IIoT)
- Robot-as-a-Service (RaaS) with IoT – YOI Robotics

LANGUAGES

- Tamil (Native)
- English (Professional Working Proficiency)
- Hindi (Limited Working Proficiency)